

Score: _____ / **10**

Calculus I (MATH 2200)

Spring 2020

Wednesday, 02/05/2020

Quiz 2: Section 2.3, 2.4

Name (Print): _____

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Section 2.3

1. Given that $\lim_{x \rightarrow 2} f(x) = 1$ and $\lim_{x \rightarrow 2} g(x) = -11$, find the indicated limits.

2 pts

(a) $\lim_{x \rightarrow 2} 2g(x) + 5f(x)$

2 pts

(b) $\lim_{x \rightarrow 2} \frac{2x - f(x)}{x^2 + 11}$

- 2 pts 2. Suppose f satisfies $2 \leq f(x) \leq 1 + x^2$ for all values of x near 1. Find $\lim_{x \rightarrow 1} f(x)$.

- 2 pts 3. Let $f(x) = \frac{|x - 2|}{x - 2}$. Explain why the limit $\lim_{x \rightarrow 2} f(x)$ does not exist.

Section 2.4

2 pts 1. Analyze $\lim_{x \rightarrow -4^+} \frac{-x^3 + 5x^2 - 6x}{-x^3 - 4x^2}$.

2 (bonus) 2. Analyze $\lim_{x \rightarrow 0^+} \frac{x + 5}{x}$.

1 (bonus) 3. Analyze $\lim_{x \rightarrow 0^+} \frac{1}{x}$.